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CS 4092 – Database Design / Development

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Requirements Document – Library Database

The library database should support 2 main user roles:

- Librarians: Responsible for checking out/returning books and movies, responsible for leading community activities, and responsible for maintaining the records/organization of the books/movies/CDs.
- Visitors: Can check out and return books and movies, and can attend community events.

There will be the following strong entities:

- Librarian
 - Librarian ID
 - Name
 - Address (Street, City, State, Zip Code)
 - DOB
 - Derived: age
 - Salary
 - Email (multi-value)
- Visitor
 - Visitor ID
 - Name
 - Address (Street, City, State, Zip Code)
 - DOB
 - Derived: age
 - Email (multi-value)
 - Fines
- Book
 - Book Call Number (Library of Congress Call Number)
 - Title
 - Author
 - Publishing Year
 - Status (checked out, to be shelved, available, missing)

- Location (east shelves, west shelves, basement shelves)
- Movie
 - Movie Call Number (Library of Congress Call Number)
 - Title
 - Director
 - Release Year
 - Status (checked out, to be shelved, available, missing)
 - Location (east shelves, west shelves, basement shelves)
- Community Activities
 - Community Activity ID
 - Title
 - Description
 - Time
 - Date
 - Location
 - Librarian ID (Librarian leading the activity)
 - Capacity
 - Demographic

There will be the following relationships:

- Customer checks out a book (Librarian checks them out)
 - Also checks out Movie
- Customer returns a book (Librarian checks them in)
 - Also returns Movie
- Customer attends a Community Activity led by a Librarian
- Books have a record of Customers who have checked them out previously
- Movies have a record of Customers who have checked them out previously